

CASA EFDEN



Visual: project identity graphic generated from the supplied project list. Official project images are available through the source links below.

Why is this project a best practice?

Challenge

The challenge was to turn advanced energy concepts into a full-scale, observable building that could be tested under Romania's continental climate, while remaining understandable to students, professionals and the public. The project also had to coordinate architecture, envelope, renewable systems, controls and user behaviour as one performance system.

Solution

The response combines a greenhouse used simultaneously for urban food production, daylight, indoor air quality and passive solar heating; prefabrication and an integrated architecture-engineering design developed for an international performance competition; and a public-facing living laboratory that links high-efficiency systems, healthy materials and user education. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The project is highly replicable as a method: establish a measurable energy target, prototype the envelope and systems, monitor real operation, and use the results for training and design improvement. Individual technologies can be transferred to schools, offices and homes, although sizing and control must be recalculated for each climate, occupancy pattern and budget.

Key facts and figures

Location	Bucharest, Romania (Technical University of Civil Engineering campus)
Building use / function	Solar-house prototype, education and demonstration centre

Project type (new build, refurbishment, extension)	Prefabricated demonstrator; assembled for Solar Decathlon Europe 2014 and reconstructed in Bucharest
Plot area / Floor area	Not consistently disclosed in the consulted public sources
Year / phase	Designed and exhibited in 2014; subsequently operated as a demonstration house
Climate zone	Temperate continental
Thematic focus / innovation	Solar Decathlon prototype; integrated passive and active systems; greenhouse; healthy materials; public education
Investor / building owner	EFdeN / university-led non-profit initiative
Architect	EFdeN multidisciplinary student and professional team
Other involved stakeholders	Technical University of Civil Engineering Bucharest; students, academics, industry partners; Romania Green Building Council
Contact person	EFdeN - https://efden.org/
Further information	https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf https://efden.org/

Casa EFdeN is a prefabricated demonstrator; assembled for solar decathlon europe 2014 and reconstructed in bucharest project in Bucharest, Romania (Technical University of Civil Engineering campus). It functions as solar-house prototype, education and demonstration centre. Its main best-practice contribution is A student-led Romanian Solar Decathlon house that became a permanent education and innovation platform. The documented strategy brings together a greenhouse used simultaneously for urban food production, daylight, indoor air quality and passive solar heating, prefabrication and an integrated architecture-engineering design developed for an international performance competition, and a public-facing living laboratory that links high-efficiency systems, healthy materials and user education. The RoGBC toolkit confirms Solar Decathlon Europe 2014 participation and Green Homes certification; detailed performance values vary by competition and later operating configuration. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - A greenhouse used simultaneously for urban food production, daylight, indoor air quality and passive solar heating.
2. Integrated systems and operation - Prefabrication and an integrated architecture-engineering design developed for an international performance competition.
3. Wider value - A public-facing living laboratory that links high-efficiency systems, healthy materials and user education.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A

verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.